



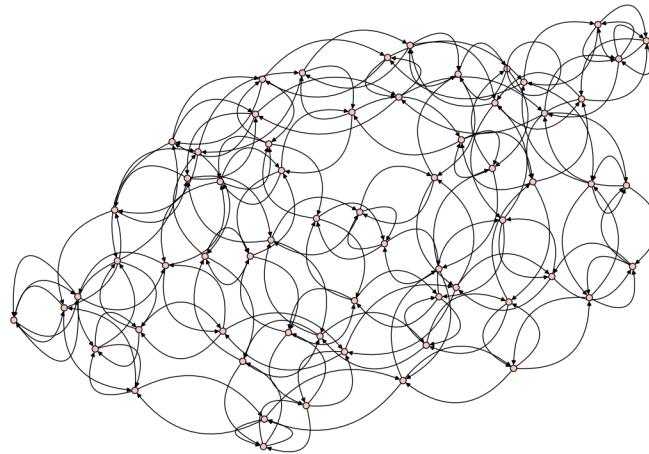
THE UNIVERSITY OF
MELBOURNE

FINAL REPORT

The Art of Scientific Computation: Spectral and Mixing Properties of Supersingular Isogeny Graphs

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Chapter 1

Introduction

1.1 Overview

In this project, I will construct and analyse supersingular ℓ -isogeny graphs over finite fields $\mathbb{F}_{p^2}$. These are graphs whose vertices represent isomorphism classes of supersingular elliptic curves over $\mathbb{F}_{p^2}$ and whose edges represent degree- ℓ isogenies between them (see [2] for background). These graphs are of special interest in cryptography, since many cryptographic protocols can be understood as applying random walks on these graphs. One such example is given in [1], whose designs provide inspiration for the graphs considered in this project.

The significance of these graphs in cryptography arises mainly from their strong expansion properties; in particular, they are Ramanujan graphs, and therefore random walks on these graphs converge quickly to a uniform distribution. The most cryptographically interesting consequence is that given any two vertices in one such graph, it will be computationally expensive to find a path connecting them.

We are therefore interested in performing empirical verification of theoretical properties attributed to these graphs. To this end, two main interests stand out: the eigenvalue spectra associated to the adjacency matrices of these graphs, and the mixing times of random walks on these graphs. For this project, I will produce a sample of supersingular isogeny graphs over a range of parameters and compute these two characteristics, comparing them to theoretical estimates in order to illustrate and corroborate them.

1.2 Scope

We have the following primary objectives.

1. Efficiently compute supersingular ℓ -isogeny graphs over $\mathbb{F}_{p^2}$ for a range of parameters p and ℓ .
2. Compute the adjacency matrices for a sample of supersingular isogeny graphs and compute their eigenvalue spectra. We will use this to answer two questions.
 - (a) How does the spectral gap of a supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$ vary according to the parameters ℓ and p ?
 - (b) How does the spectral gap of a supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$, as measured empirically, align with theoretical estimates of the same quantity?
3. Produce, for each sampled supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$, a sample of random walks on that graph. We will use this to answer two questions.
 - (a) How are mixing times of random walks distributed for each graph, in terms of parameters ℓ and p ?
 - (b) How do the mixing times of random walks on these graphs, as measured empirically, align with theoretical estimates of mixing times for Ramanujan graphs?

Chapter 2

Mathematical Background

2.1 Elliptic curves

The definition of elliptic curve can be rendered in various equivalent forms. For instance, as per [3], we could define an elliptic curve E over a field K as a non-singular projective algebraic curve of genus 1 having a distinguished K -rational point $O \in E$. But, we are interested in elliptic curves with a view to computation, and therefore give a simpler definition, more tractable for our purposes.

Definition 1 (Elliptic Curve). Let K be a field of characteristic other than 2 or 3. Then an elliptic curve E over K consists of the set of solutions $(x, y) \in K^2$ of the short Weierstrass equation

$$E : y^2 = x^3 + ax + b$$

where $a, b \in K$ and $4a^3 + 27b^2 \neq 0$.

Over the real numbers $\mathbb{R}$, elliptic curves can be viewed as familiar planar curves.

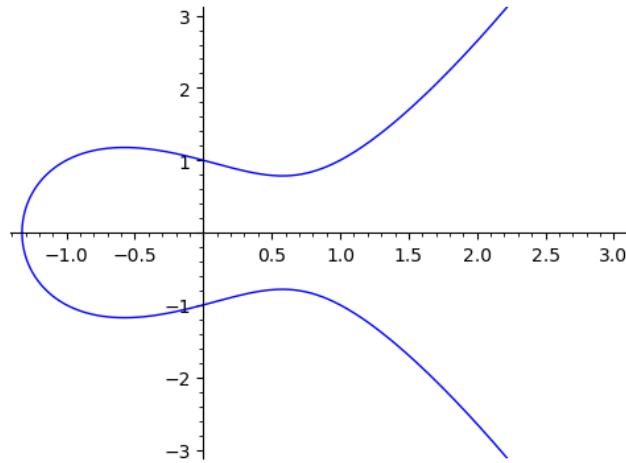


Figure 2.1: Elliptic curve $E : y^2 = x^3 - x + 1$ over $\mathbb{R}$.

However, we will be mainly interested in elliptic curves over finite fields $\mathbb{F}_{p^n}$ for prime p and $n \in \mathbb{Z}_{\geq 1}$. An example of the solution set to one such curve over a finite field is shown below.

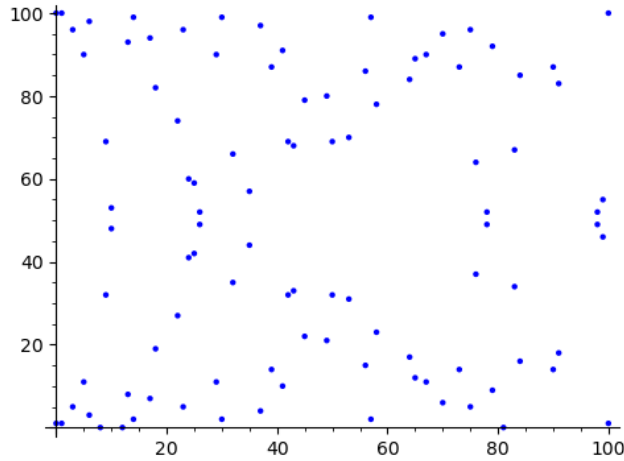


Figure 2.2: Elliptic curve $E : y^2 = x^3 - x + 1$ over $\mathbb{F}_{101}$, the finite field of order 101.

Key to the theory of elliptic curves is the fact that it is possible to define a group operation (conventionally called a *group law*) on the points of E , satisfying certain desirable properties. The definition of this operation arises from projective geometry, whose details we defer to a reference such as [3]. For our purposes, it will suffice to understand that any elliptic curve has a group structure.

Definition 2 (Group Law). Let E be an elliptic curve over a field K . It is a consequence of Bézout’s theorem that under mild assumptions and counting multiplicity, an algebraic curve of degree n and an algebraic curve of degree m will intersect in nm points. Since an elliptic curve has degree 3 and a line has degree 1, it follows that the line spanned by two points $P, Q \in E$ will intersect E at a unique third point $R' \in E$. We let $R \in E$ be the reflection of R' over the x -axis and define a *group law* $P \oplus Q = R$, such that $(E, \oplus)$ forms an abelian group. (For details, see [3].)

STILL IN PROGRESS

2.2 Isogenies

Definition of elliptic curve isogenies, the degree of an isogeny, the relationship between an isogeny and its kernel, and computational aspects such as Vélu’s formulas.

2.3 Supersingular ℓ -isogeny graphs

Definition of supersingular isogeny graphs. Some discussion of their spectral and mixing properties as described in existing literature.

2.4 Basic spectral graph theory

Basic notions of spectral graph theory, particularly the spectral gap.

2.5 Random walks and mixing times

Introduction to the notions of random walks and their mixing times.

Chapter 3

Computational Implementation

I haven't quite gotten around to writing this up but I promise I do have working code. Attached are some sample supersingular isogeny graphs I have produced; I have begun analysing these but have yet to write out my results.

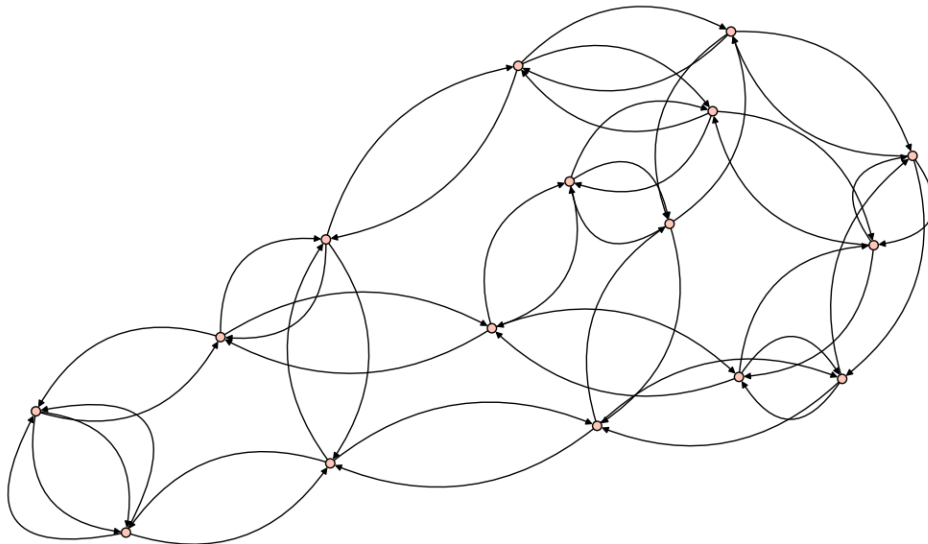


Figure 3.1: Supersingular 2-isogeny graph over the finite field F_{193^2} .

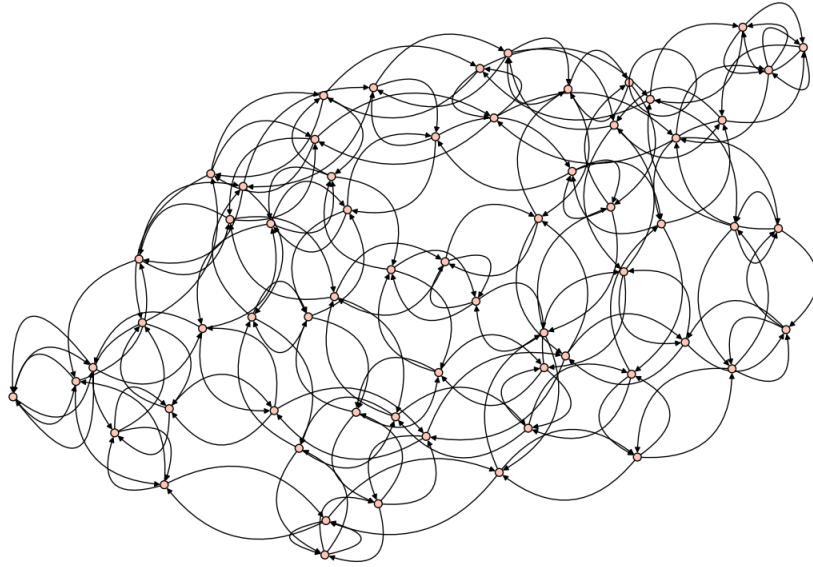


Figure 3.2: Supersingular 2-isogeny graph over the finite field F_{757^2} .

3.1 Elliptic curves in Sage

Introduction to the Sage software package and its facilities for elliptic curve operations.

3.2 Generating ℓ -isogeny graphs

3.3 Computing eigenvalue spectra

3.4 Generating random walks

3.5 Computing mixing times

Chapter 4

Results

Discuss the results of computational experiments, produce plots of results against varying parameters p and ℓ . Produce visualisations of small isogeny graphs.

Chapter 5

Discussion

Compare results from previous chapter with known results in existing literature. Especially compare asymptotic estimates found in literature with the empirical results we find.

Chapter 6

Conclusion

Summarise findings, suggest further possible questions.

Appendix A

Code Excerpts

Will include any especially interesting or relevant pieces of code. Full codebase will be available at a linked GitHub repository.

Bibliography

- [1] W. Castryck, T. Lange, C. Martindale, L. Panny, and J. Renes. Csidh: An efficient post-quantum commutative group action. In *Advances in Cryptology – ASIACRYPT 2018: 24th International Conference on the Theory and Application of Cryptology and Information Security, Brisbane, QLD, Australia, December 2–6, 2018, Proceedings, Part III*, page 395–427, Berlin, Heidelberg, 2018. Springer-Verlag.
- [2] L. Panny. Elliptic curves and isogenies: The good bits, July 2021. Isogeny-based Cryptography School Week 1 notes, Bristol.
- [3] J. H. Silverman. *The Arithmetic of Elliptic Curves*. Graduate Texts in Mathematics. Springer New York, NY, 2nd edition, 2009.